

conv-file-quality-assurance

conv-file-quality-assurance is a cross-platform application designed to verify the integrity and quality of files that have been converted from their original format to archival formats. It performs a series of in-depth comparisons and checks, such as visual differences, metadata mismatches, and structural inconsistencies, to ensure the preservation process retains fidelity.

HomeSettingsReportError Analysis

Original file Folder

Select

New file Folder

Select

☐ Start from checkpoint

Select

Data Extraction Folder

Views

Extract

Quick Settings

1

Enable Size Comparison

✓

1

Enable Resolution Comparison

✓

1

Enable Color Profile Comparison

✓

1

Enable Font Comparison

✓

1

Enable Point by Point Comparison

✓

1

Enable Visual Document Comparison

✓

1

Enable Page Count Comparison

✓

1

Enable Animation Comparison

✓

1

Enable Table Break Check

✓

0%

Start

Stop

Table of contents

- [Background & Motivation](#)
- [Key Features](#)
- [File pairing](#)
- [Comparison Methods](#)
- [Screenshots & UI Overview](#)
- [Supported File Formats](#)
 - [Comparison Methods in Supported Pipelines](#)
- [Installation](#)
 - [Windows](#)
 - [Ubuntu 22.04](#)
- [Usage Guide](#)
- [Known Issues & Limitations](#)
- [Testing & Validation](#)
- [License](#)
- [Third-Party Licenses](#)
- [Contributing](#)

Background

This project is a proposal from Innlandet Fylkeskommune for a bachelor thesis at the Norwegian University of Science and Technology's programming program. It addresses a key issue in the automatic conversion of documents to archival formats—namely, the potential for errors and inconsistencies during conversion.

The Innlandet County Archive, one of Norway's largest inter-municipal archives, receives a wide variety of digital documents from 48 municipalities. Many of these documents arrive in formats not suited for long-term archival storage, requiring conversion to approved archival formats.

A previous NTNU thesis developed an automated conversion tool. However, during conversion, loss of data or visual discrepancies can occur. Because of the large volume of files, manual inspection is not feasible.

To solve this, the archive requested the development of a quality assurance tool that can automatically detect potential issues in converted files. This project aims to fulfill that need—reducing manual work and increasing reliability in the archival process.

Key features

- Pixel-by-pixel visual comparisons of images or document pages
 - Detection of metadata differences (resolution, transparency, color space, bit depth, etc.)
 - Table break detection in spreadsheets converted to PDFs
 - Animation detection in presentation formats (PPTX, ODP)
 - Font comparison
 - PDF page count comparisons
 - Extraction and verification of embedded images
 - JSON report generation with severity-based error classification
-

File Pairing

To perform comparisons, the application pairs files from the original and converted directories based on their filenames.

- If a file in the original directory has a unique name, its converted counterpart keeps the same name.
 - If there are multiple original files with the same name (but different extensions), the converted files are renamed by appending the original extension (e.g., document.docx → document_DOCX.pdf) to ensure unique and traceable pairings.
-

Comparison Methods

Each method performs a specific type of comparison between original and converted files:

- **File Size** – Compares raw file sizes.
 - **Pixel-by-Pixel** – Calculates Euclidean distance between color values for each pixel.
 - **Color Space** – Identifies missing or altered color spaces.
 - **Fonts** – Flags missing or replaced fonts, as well as text colors and background colors.
 - **Number of Pages** – Detects discrepancies in page count.
 - **Animations** – Flags the presence or absence of animations in presentation files.
 - **Image Resolution** – Compares DPI and resolution metadata.
 - **Visual Document Comparison** – Hybrid method: compares document layout + pixel comparisons.
 - **Table Break Check** – Flags if tables/images are split across multiple pages in PDFs.
 - **Metadata Comparison** – Compares physical dimensions, bit depth, resolution, etc.
 - **Extracted Metadata Check** – Extracts images from documents and checks embedded metadata
- NOTE: This method may trigger frequently; severity could be reduced.*
-

Screenshots & UI Overview

Home Screen

HomeSettingsReportError Analysis

Original file Folder

Select

New file Folder

Select

☐ Start from checkpoint

Select

Data Extraction Folder

Views

Extract

Quick Settings

☒ Enable Size Comparison

☒

☒ Enable Resolution Comparison

☒

☒ Enable Color Profile Comparison

☒

☒ Enable Font Comparison

☒

☒ Enable Point by Point Comparison

☒

☒ Enable Visual Document Comparison

☒

☒ Enable Page Count Comparison

☒

☒ Enable Animation Comparison

☒

☒ Enable Table Break Check

☒

0%

Start

Stop

Settings Tab

Customize which checks are performed.

HomeSettingsReportError Analysis

Reset Settings

Save Settings

Default ▾

General

File Formats

☒ Size Comparison

☒ Enable Size Comparison

Comparison Threshold:

75,00

 %

☒ Image Resolution Comparison

☒ Color Profile Comparison

☒ Font Comparison

☒ Point By Point Comparison

☒ Visual Document Comparison

☒ Page Count Comparison

☒ Animation Comparison

Report Tab

Visual interface for exploring the generated JSON reports.

HomeSettingsReportError Analysis

Load current reportLoad from JSONClear

10/12 file comparisons failed | Completed in 00:00:05

Search

Show failed comparisons firstShow passed comparisons first

No ignored files.

FAIL - 13_f-fc.rtf → 13_f-fc.pdf (fmt/355 → fmt/477)

FAIL - 5_f-fc.odt → 5_f-fc.pdf (fmt/290 → fmt/477)

Passed: No
Tests (3/5 passed)

FAIL - Fonts

PASS - Pages

FAIL - Size

Passed: No
Errors:

File Size Difference: The difference in size for the two files exceeds expected values.

Error Analysis Tab

Get a summary of test failures by severity.

HomeSettingsReportError Analysis

Load From current reportLoad from JSONClear

18 Tests failed

Size (6)

Fonts (6)

13_f-fc.rtf → 13_f-fc.pdf
5_f-fc.odt → 5_f-fc.pdf
2_f-fc.docx → 2_f-fc.pdf
8_f-fc.xlsx → 8_f-fc.pdf
1_f.pptx → 1_f.pdf
4_f.odp → 4_f.pdf

Color profile (2)

Metadata (1)

Transparency (2)

Table Break Check (1)

Supported File Formats

Format	JPEG	PNG	GIF	TIFF	BMP	DOCX	PPTX	XLSX	ODT	ODP	ODS	PDF/1.0-1.6	PDF/A1-4	XML	EML	CSV	RTF	HTML
JPEG	✓	✓	✓	✓	✓							✓	✓					
PNG	✓	✓	✓	✓	✓							✓	✓					
GIF	✓	✓	✓	✓	✓							✓	✓					
TIFF	✓	✓	✓	✓	✓							✓	✓					
BMP	✓	✓	✓	✓	✓							✓	✓					
DOCX						✓			✓			✓	✓				✓	
PPTX							✓			✓		✓	✓					
XLSX								✓			✓	✓	✓					
ODT						✓			✓			✓	✓				✓	
ODP							✓			✓		✓	✓					
ODS								✓			✓	✓	✓					
PDF/1.0-1.6												✓	✓					
PDF/A1-4												✓	✓					
XML												✓	✓					
EML												✓	✓					
CSV												✓	✓					
RTF						✓			✓			✓	✓					
HTML												✓	✓					

🔍 Comparison Methods in Supported Pipelines

The comparison is performed by functions called pipelines. Pipelines compare the original file to the converted file. In this list they follow the [original_format]-[converted_format] name schema.

- CSV-PDF
 - File Size Comparison
 - Table break Checks
- DOCX-PDF&TextDoc*
 - Font Comparison
 - Page Count Comparison
 - File Size Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
- EML-PDF
 - Font Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
- HTML-PDF
 - Font Comparison
- IMG-IMG**
 - File Size Comparison
 - Resolution Comparison
 - Image Metadata Comparison
 - Pixel Value Comparison
 - Color Profile Comparison
- IMG-PDF**
 - File Size Comparison
 - Resolution Comparison
 - Image Metadata Comparison
 - Pixel Value Comparison
 - Color Profile Comparison
- ODP-PDF
 - Font Comparison
 - Page (slide) Count Comparison
 - File Size Comparison
 - Animation Check
 - Color Profile Comparison (extracted)

- Image Metadata Comparison (extracted)
- ODP-Presentation***
 - Font Comparison
 - Page (slide) Count Comparison
 - File Size Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
- ODS-PDF
 - Font Comparison
 - File Size Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
 - Table Break Check
 - Check for blank pages at the end
- ODS-Spreadsheet****
 - Font Comparison
 - File Size Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
- ODT-PDF&TextDoc*
 - Font Comparison
 - Page Count Comparison
 - File Size Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
- PDF-PDF
 - Font Comparison
 - Page Count Comparison
 - File Size Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
 - Visual Document Comparison
- PDF-TextDoc*
 - Font Comparison
 - Page Count Comparison
 - File Size Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
- PPTX-PDF
 - Font Comparison
 - Page Count Comparison
 - File Size Comparison
 - Animation Check
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
- PPTX-Presentation***
 - Font Comparison
 - Page (slide) Count Comparison
 - File Size Comparison
 - Color Profile Comparison (extracted)
 - Image Metadata Comparison (extracted)
- RTF-PDF&TextDoc*
 - Font Comparison
 - File Size Comparison
 - Color Profile Comparison
 - Page Count Comparison

- o Color Profile Comparison (extracted)
 - o Image Metadata Comparison (extracted)
- XLSX-PDF
 - o Font Comparison
 - o File Size Comparison
 - o Color Profile Comparison (extracted)
 - o Image Metadata Comparison (extracted)
 - o Table Break Check
 - o Check for blank pages at the end
- XLSX-Spreadsheet****
 - o Font Comparison
 - o File Size Comparison
 - o Color Profile Comparison (extracted)
 - o Image Metadata Comparison (extracted)

* TextDoc refers to RTF, ODT and DOCX formats

** IMG refers to PNG, JPEG, BMP, TIFF and GIF formats

*** Presentation refers to PPTX and ODP

**** Spreadsheet refers to ODS and XLSX

Installation

Windows

Dependencies

- [.NET 8+ \(https://dotnet.microsoft.com/en-us/download\)](https://dotnet.microsoft.com/en-us/download)
- [Siegfried \(https://www.itforarchivists.com/\)](https://www.itforarchivists.com/) (Add to system PATH)

Installation Steps

```
git clone --recursive https://github.com/panmat7/conv-file-quality-assurance.git
cd conv-file-quality-assurance
dotnet build
```

Ubuntu 22.04

Dependencies

- .NET 8+
- [Siegfried \(https://www.itforarchivists.com/\)](https://www.itforarchivists.com/)
- [Emgu.CV \(https://www.emgu.com/wiki/index.php/Download_And_Installation\)](https://www.emgu.com/wiki/index.php/Download_And_Installation)
- [ExifTool \(https://exiftool.org/install.html#Unix\)](https://exiftool.org/install.html#Unix)

Installation Steps

1. **Install Siegfried**
Download and install from: <https://www.itforarchivists.com/>
2. **Install Emgu.CV**
Follow instructions here: https://www.emgu.com/wiki/index.php/Download_And_Installation
 - o If you can't run the Point by point or Visual document comparison, it is probably because of missing dependencies. Check the provided resource and follow the instructions (https://www.emgu.com/wiki/index.php/Download_And_Installation#System.DllNotFoundException)

3. **Install ExifTool**

Using apt:

```
sudo apt-get install libimage-exiftool-perl
```

Or follow the instructions at: <https://exiftool.org/install.html#Unix>

4. **Install .NET 8 SDK**

5. **Clone and Build the Project**

```
git clone --recursive https://github.com/panmat7/conv-file-quality-assurance.git
cd conv-file-quality-assurance
dotnet build
```

Usage Guide

1. **Choose original and converted folders.**
To extract metadata only, select a single directory and press **"Extract"**.
2. **Click "Load"** to pair original and converted files.
Optionally load from a **checkpoint JSON** to resume interrupted runs.
 - To do this, check **"Start from Checkpoint"** and select a saved checkpoint JSON report.
3. **Configure checks** using the **Settings** tab or **Quick Settings**.
 - You can also exclude certain file formats from verification in the Settings tab.
4. **Click "Start"** to begin the verification process.
5. Once complete, a JSON report will be saved in the `reports/` folder (created in the working directory).
6. **View the report:**
 - Go to the **"Report"** tab.
 - Click **"Load from JSON"** and select the report to view results.
7. **Error Analysis Tab:**
 - Navigate to the **"Error Analysis"** tab to view a general overview of test failures by severity.

⌚ **Note:** The console prints progress every 5 minutes with estimated time remaining.

⚠ **Severity Levels:** Errors are categorized as **Low**, **Medium**, or **High**.

- **Low** should be considered warnings and not always critical.

Known Issues & Limitations

- **Table break check** currently works only for PDFs with standard A4 or letter-sized pages.
- **Extracted metadata comparison** may over-report differences. We are evaluating lowering the severity of these alerts.
- Some Emgu.CV-based checks may not function if system dependencies are missing (especially on Linux).

Testing & Validation

- Application manually tested on:
 - **Windows 10 & 11**
 - **Ubuntu 22.04**
- JSON output reports pass internal consistency checks.
- Planned: Schema validation for report files to ensure correctness.

License

This project is licensed under the [GNU GENERAL PUBLIC LICENSE v3.0 \(LICENSE\)](#).

Third-Party Licenses

This project uses external libraries such as Emgu.CV, ExifTool, and Siegfried.
See the [NOTICE \(NOTICE\)](#) file for third-party license information.

Contributing

We are not accepting contributions for now.